

Quiz - 2
Section - 3J
Date - 18-Sept-25

Student Roll No. [REDACTED]
Student Name: [REDACTED]
Total marks: 10

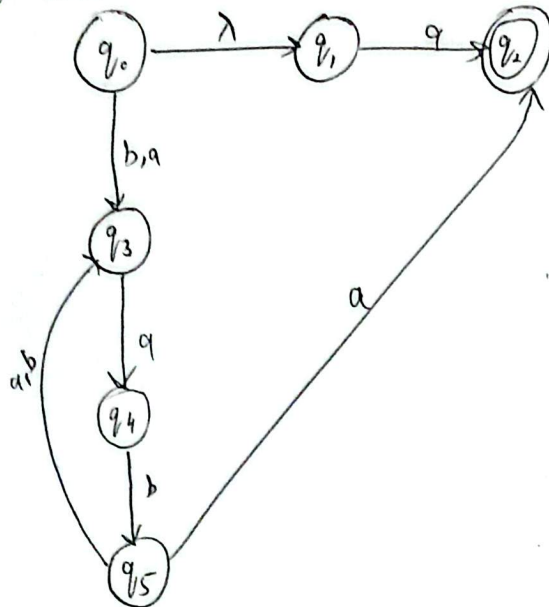
Q.5

Note: In both questions, do all steps neatly in sequence. Otherwise, no marks will be given.

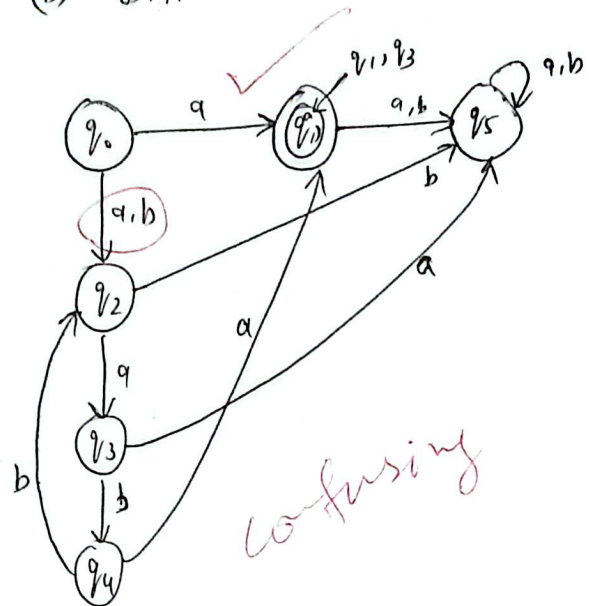
Q:1 - Given a R.E. $\{bab, aab\}^*.a$ when $\Sigma = \{a, b\}$ (2+2+1 marks)

- Draw NFA – state diagram
- Convert NFA of part (a) into DFA
- Is the string 'babaabbaba' acceptable by language?

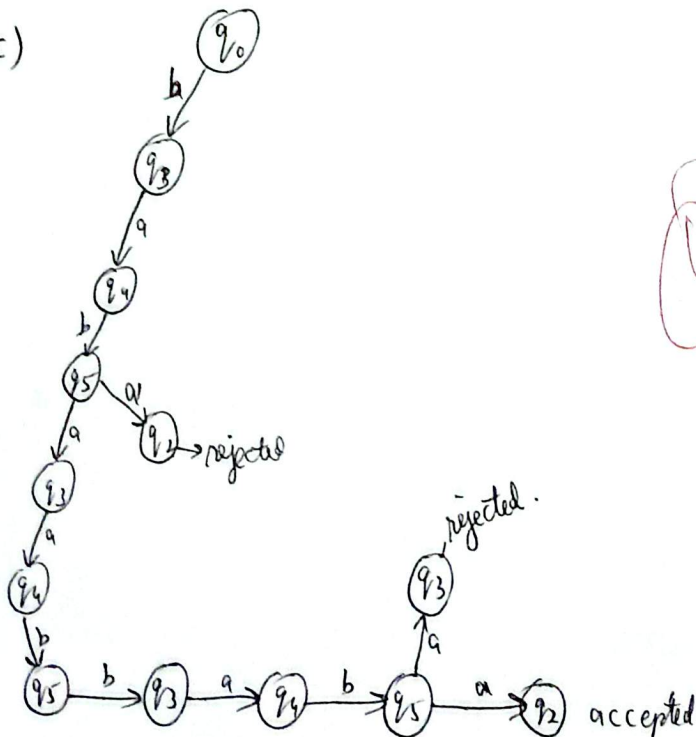
(a) NFA



(b) DFA



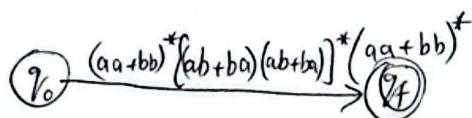
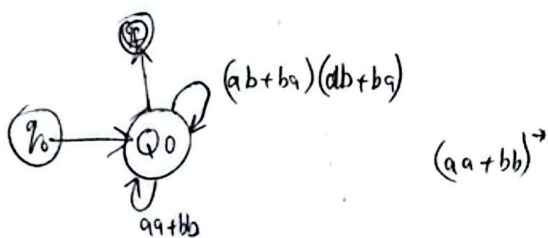
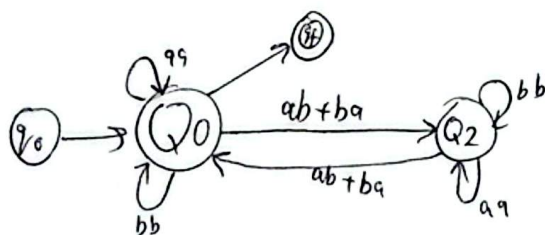
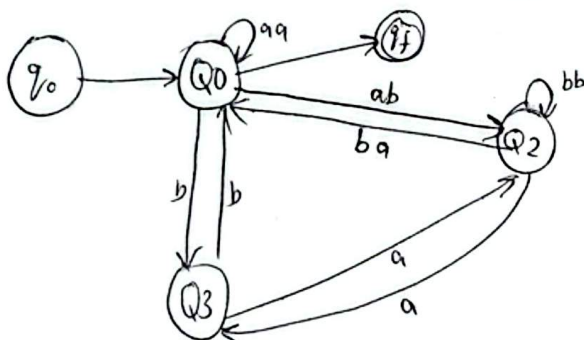
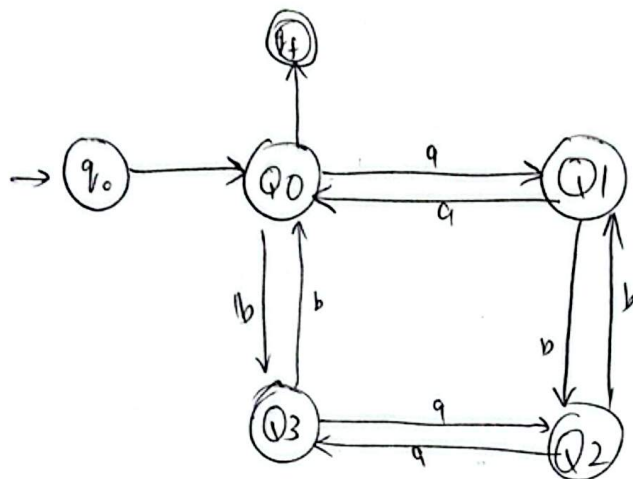
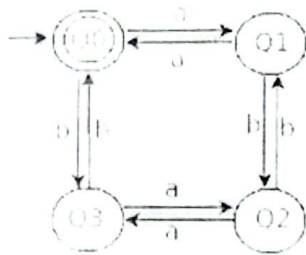
(c)



Q

Yes

Q:2 – Make Regular Expression of language accepted by the DFA below. (5 marks)
Follow state elimination in sequence Q1, Q3, Q2, Q0 to get final expression.



4.5

So, the expression is:

$$(aa+bb)^* [(ab+ba)(ab+ba)]^* (aa+bb)^*$$